UNIVERSITY OF ILLINOIS URBANA–CHAMPAIGN

Economic Forecasting Final Project

ECON 475 · Summer 2026

Answers and supporting empirical results

Three forecasting applications

01 Monthly U.S. residential electricity retail sales

02 Monthly seasonally adjusted WTI crude-oil prices

03 Daily Disney stock prices and conditional volatility

August 07, 2026

Report guide

Scope. All estimates use the training samples specified in the project. Model selection follows the course rule of choosing the lowest BIC among the requested candidates.

Reproducibility. Every table, number, interval, and figure in this document is generated by the accompanying ECON475_FinalProject.R file. The R code is submitted separately and is not reproduced in this answer document.

Interpretation. Each part states the method, reports the relevant output, interprets it, and gives a direct conclusion. Forecast intervals are 95 percent intervals.

Question	Training sample	Forecast / test sample
1 · Electricity	1973M01–2010M12	2011M01–2011M12
2 · WTI oil	1986M01–2022M12	2023M01–2024M10
3 · Disney	2007-01-03–2024-09-30	2024-10-01–2024-11-13

Question 1

U.S. residential electricity retail sales

Monthly data. The dependent variable is y_t = log(elec$_t$); the training sample ends in December 2010.

(a) Log electricity retail sales

Method. Transform electricity sales to natural logs and plot the complete monthly series.

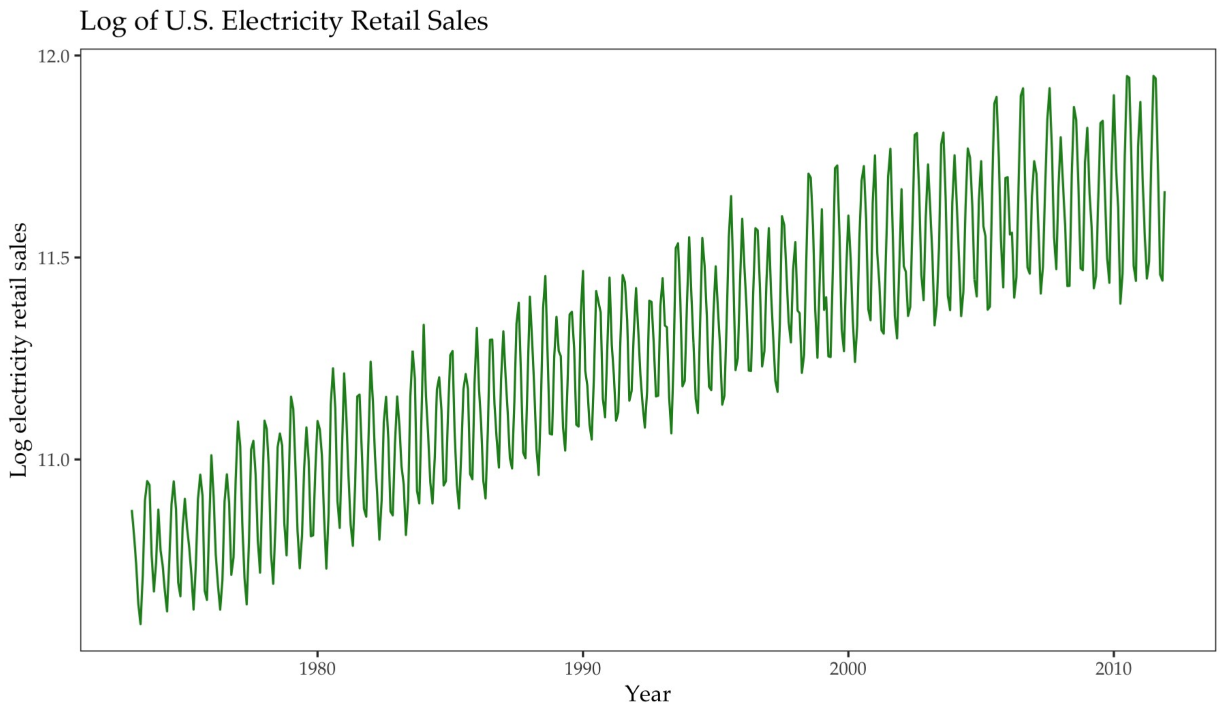

Figure 1. Log electricity retail sales, January 1973–December 2011.

Interpretation. The series has a clear upward long-run movement with curvature. It also has a pronounced repeating annual pattern, with summer and winter peaks and spring and autumn troughs.

Conclusion. A deterministic trend and monthly seasonality must be modeled before the cyclical component is analyzed.

(b) Linear versus quadratic trend

Method. Estimate linear and quadratic time-trend regressions on the training sample and compare AIC and BIC.

Model	AIC	BIC	Residual s.e.
Linear trend	-459.18	-446.81	0.1456
Quadratic trend	-463.76	-447.27	0.1447

Model	Parameter	Estimate	Std. error
Linear trend	(Intercept)	10.775594	0.013660
Linear trend	time	0.002114	5.180e-05
Quadratic trend	(Intercept)	10.736465	0.020422
Quadratic trend	time	0.002626	2.064e-04
Quadratic trend	time2	-1.122e-06	4.373e-07

Interpretation. The quadratic model has the lower BIC (−447.27 versus −446.81) and the lower AIC. Its squared-time coefficient is negative, so growth gradually flattens over the sample.

Conclusion. Choose the quadratic trend and use it in the remaining parts.

(c) Seasonal evidence in trend residuals

Method. Compute the ACF of the quadratic-trend residuals through lag 25.

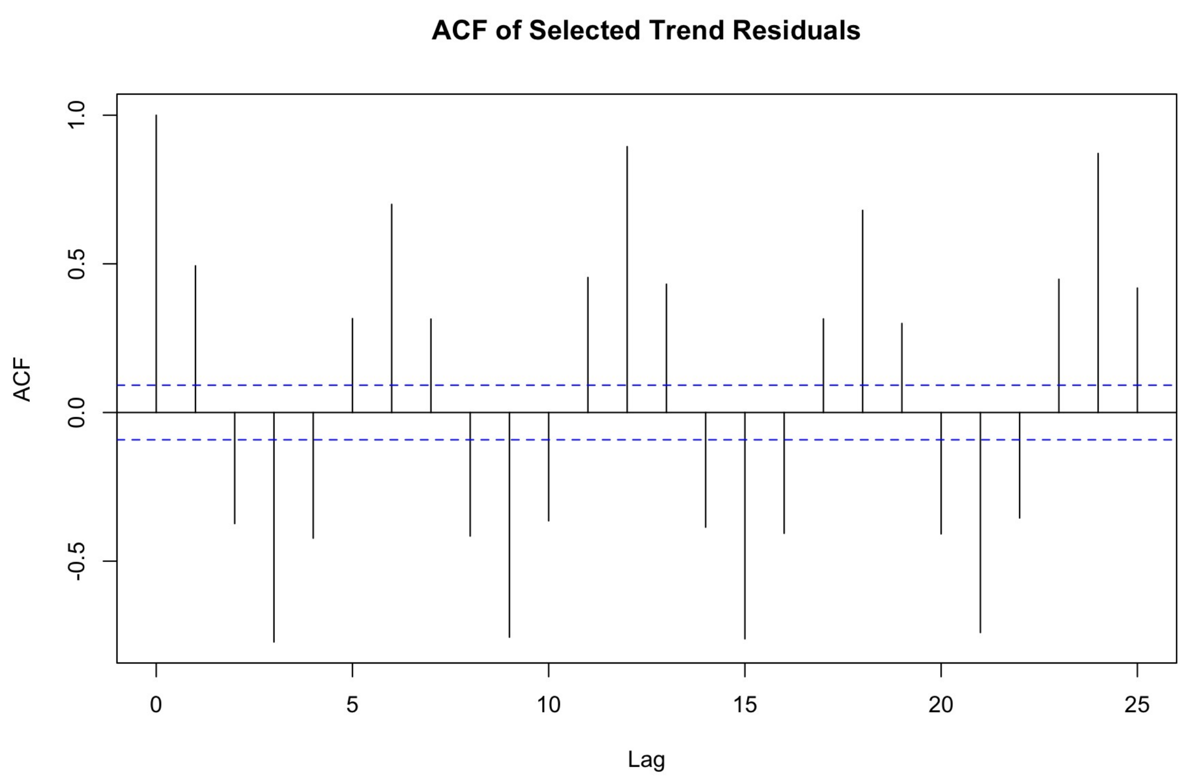

Figure 2. ACF of quadratic-trend residuals through lag 25.

Interpretation. The ACF has very large positive spikes at lags 12 (0.894) and 24 (0.871), together with an oscillating within-year pattern.

Conclusion. There is strong annual seasonality that a trend-only model does not explain.

(d) Quadratic trend with a full set of monthly dummies

Method. Estimate the selected quadratic trend jointly with twelve monthly indicators and no intercept. This is a full dummy set without a dummy-variable trap.

```
Call:
lm(formula = y ~ -1 + month + time + time2, data = q1_training)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.134293 -0.032162 -0.005265  0.030740  0.147045 

Coefficients:
             Estimate    Std. Error  t value             Pr(>|t|)    
month1  10.9111692858  0.0106923049 1020.469 < 0.0000000000000002 ***
month2  10.7894230553  0.0107000045 1008.357 < 0.0000000000000002 ***
month3  10.7018522722  0.0107075742  999.466 < 0.0000000000000002 ***
month4  10.5653203517  0.0107150140  986.030 < 0.0000000000000002 ***
month5  10.5394049786  0.0107223242  982.940 < 0.0000000000000002 ***
month6  10.7083373043  0.0107295050  998.027 < 0.0000000000000002 ***
month7  10.9074384718  0.0107365567 1015.916 < 0.0000000000000002 ***
month8  10.9321776789  0.0107434795 1017.564 < 0.0000000000000002 ***
month9  10.8069070689  0.0107502736 1005.268 < 0.0000000000000002 ***
month10 10.6099653828  0.0107569394  986.337 < 0.0000000000000002 ***
month11 10.5767227466  0.0107634771  982.649 < 0.0000000000000002 ***
month12 10.7791914751  0.0107698872 1000.864 < 0.0000000000000002 ***
time     0.0026299208  0.0000729256   36.063 < 0.0000000000000002 ***
time2   -0.0000011223  0.0000001545   -7.263     0.00000000000172 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.05114 on 442 degrees of freedom
Multiple R-squared:      1,	Adjusted R-squared:      1 
F-statistic: 1.58e+06 on 14 and 442 DF,  p-value: < 0.00000000000000022
```

Interpretation. All twelve monthly levels and both trend terms are highly significant. The month coefficients reproduce the annual pattern: the highest fitted monthly level is August, while the lowest is May. The positive linear and negative quadratic terms preserve the flattening long-run trend.

Conclusion. The deterministic component captures both trend and fixed monthly seasonality, but its residuals still require a dynamic model.

(e) Residual cycle after trend and seasonal dummies

Method. Plot the deterministic-model residuals over time and examine their ACF and PACF through lag 25.

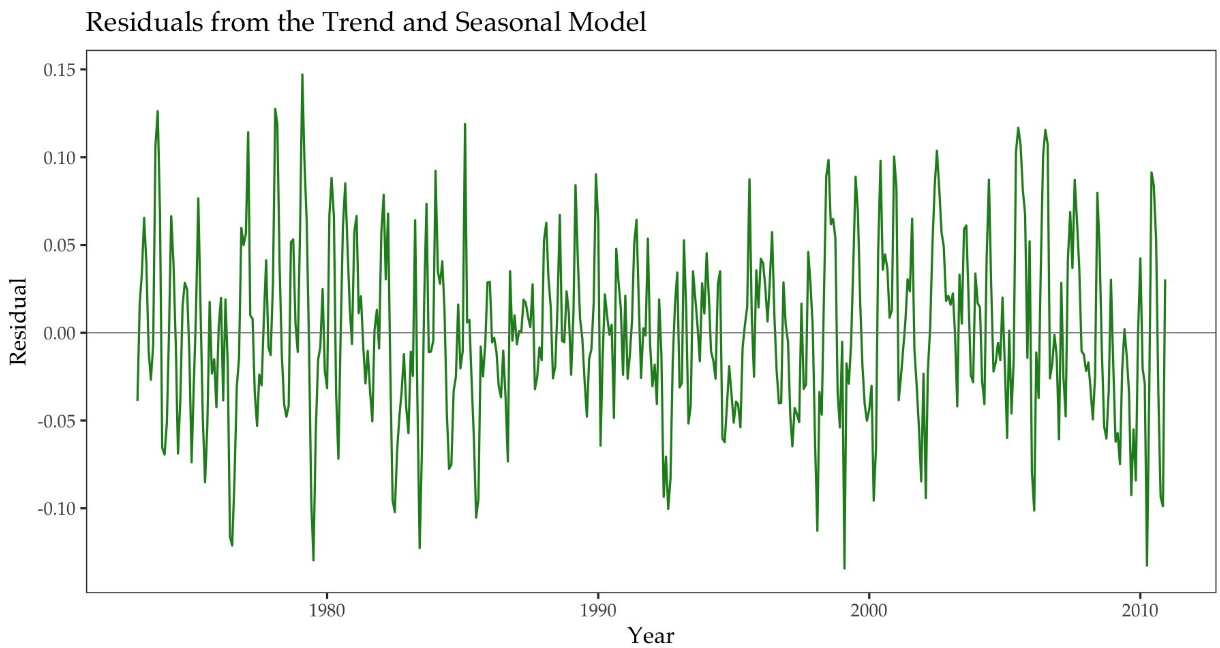

Figure 3. Residuals from the quadratic trend plus monthly dummies.

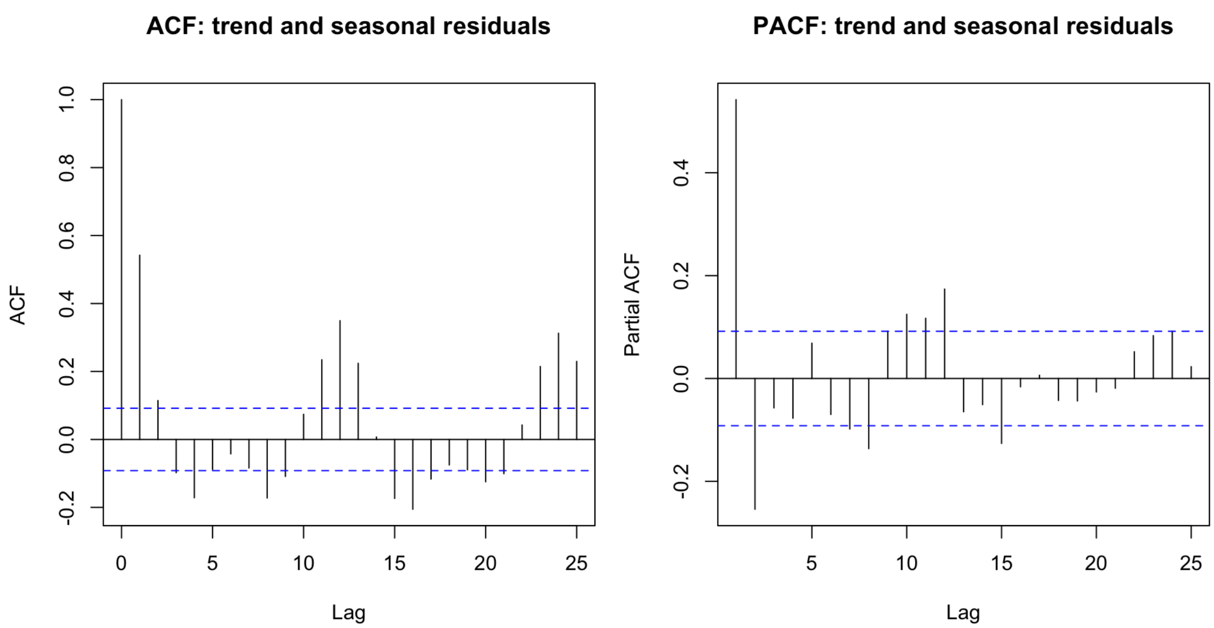

Figure 4. ACF and PACF of deterministic-model residuals.

Interpretation. The residuals remain serially correlated. The ACF is positive at lag 1 and still has visible spikes at lags 12 and 24, so fixed monthly effects do not remove the entire dynamic pattern.

Conclusion. An ARMA component is needed for the remaining cycle.

(f) ARMA model selection for the cyclical component

Method. Estimate all requested ARMA(p,q) models for p,q = 0,...,3, excluding (0,0), on the deterministic residuals. The ARMA mean is fixed at zero because the deterministic regression already includes the mean structure.

AR order	MA(0)	MA(1)	MA(2)	MA(3)
AR(0)	—	-1591.30	-1597.33	-1595.59
AR(1)	-1578.19	-1598.06	-1592.57	-1591.47
AR(2)	-1603.96	-1600.32	-1595.37	-1609.21
AR(3)	-1599.35	-1595.07	-1592.02	-1611.82

Interpretation. ARMA(3,3) has the lowest BIC, −1611.82. This specification also allows the short-run oscillation visible in the residual correlogram.

Conclusion. Choose ARMA(3,3) for the cyclical component.

(g) Selected ARMA(3,3) results and residual diagnostics

Parameter	Estimate	Std. error
ar1	1.3243	0.1194
ar2	-1.3260	0.1145
ar3	0.3346	0.1123
ma1	-0.6779	0.1252
ma2	0.7065	0.0940
ma3	0.2503	0.1091

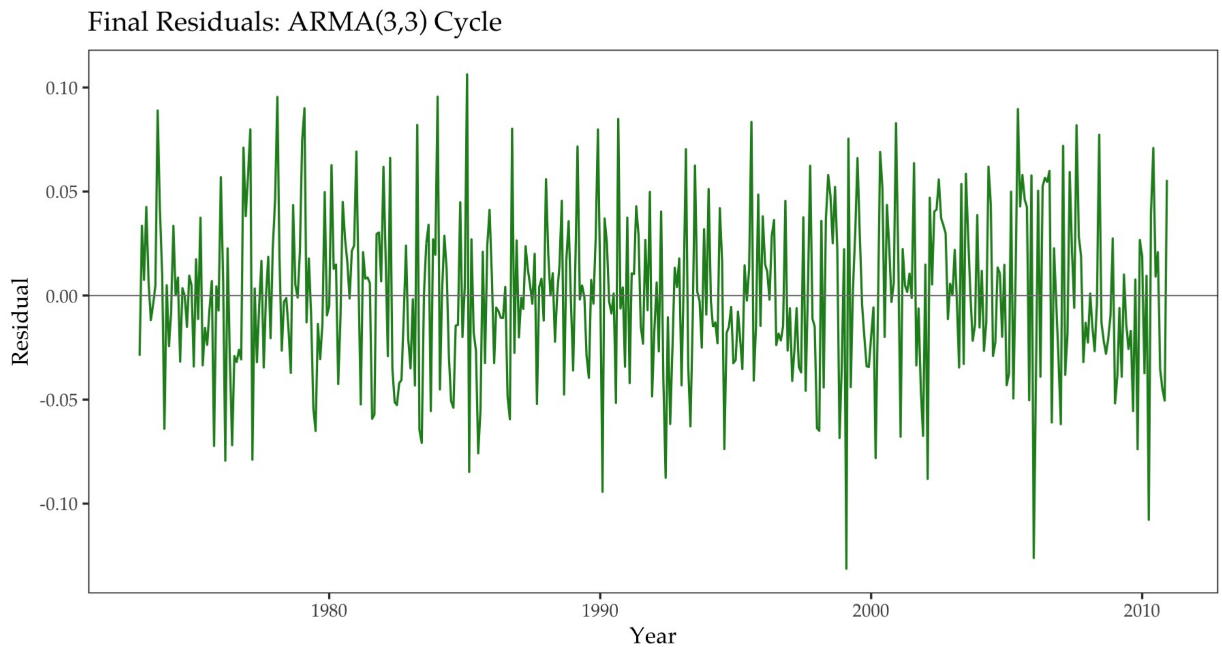

Figure 5. Residuals from the selected ARMA(3,3) cycle model.

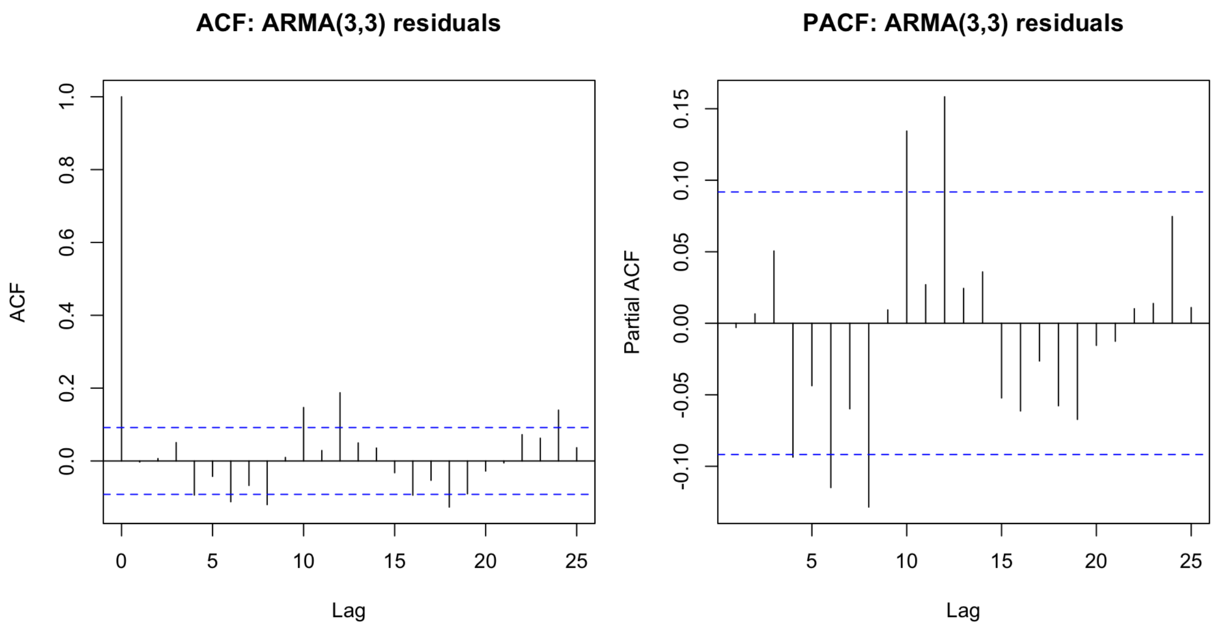

Figure 6. ACF and PACF of ARMA(3,3) residuals through lag 25.

Interpretation. The ARMA component substantially reduces the serial correlation: the lag-1 residual ACF is approximately zero. However, isolated significant spikes remain, including at seasonal lags 12 and 24. Therefore, the BIC-minimizing course model improves the residuals but does not make them perfectly white noise.

Conclusion. Retain ARMA(3,3) because it is the requested candidate with the lowest BIC, while recording the remaining residual dependence as a limitation.

(h) Forecasts for 2011

Method. Forecast the deterministic trend-seasonal component and the ARMA cycle separately, then add their point forecasts. Add the deterministic point forecast to the ARMA lower and upper limits to obtain the 95 percent interval.

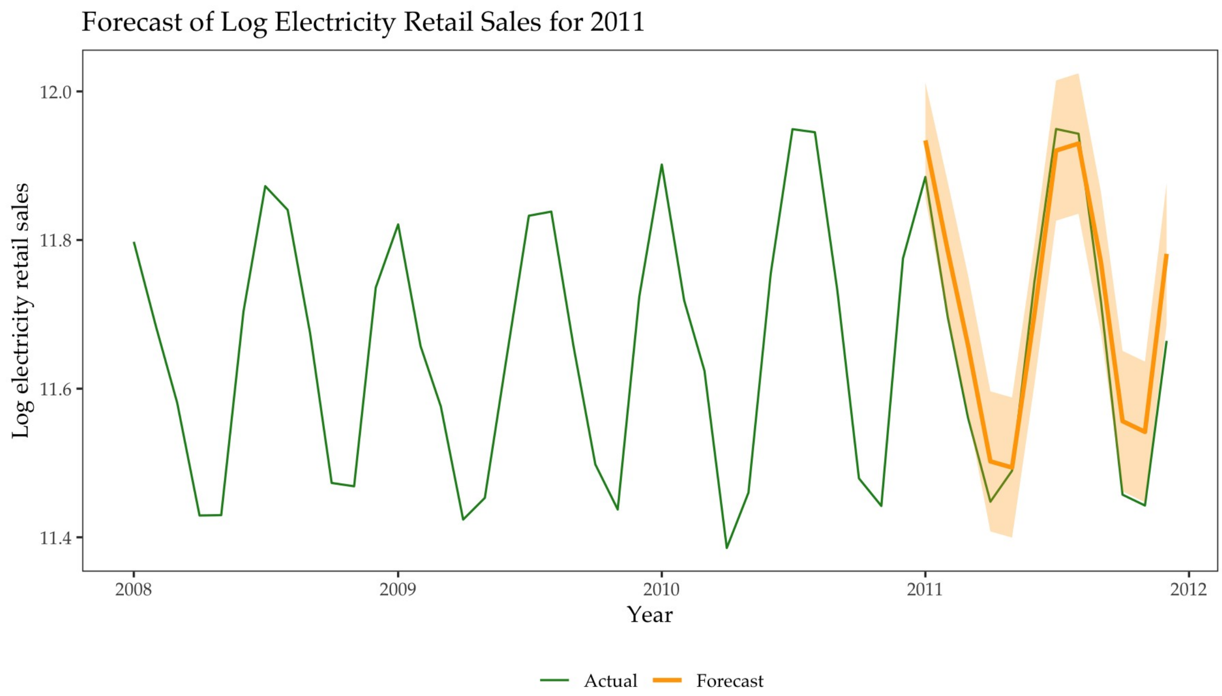

Figure 7. Actual and forecast log electricity retail sales, 2008–2011.

Month	Actual	Forecast	Lower 95%	Upper 95%
2011-01	11.885	11.934	11.857	12.012
2011-02	11.696	11.785	11.693	11.878
2011-03	11.561	11.659	11.564	11.753
2011-04	11.448	11.502	11.408	11.596
2011-05	11.490	11.494	11.400	11.588
2011-06	11.744	11.700	11.606	11.795
2011-07	11.949	11.920	11.826	12.015
2011-08	11.943	11.930	11.835	12.024
2011-09	11.718	11.770	11.676	11.865
2011-10	11.457	11.556	11.462	11.651
2011-11	11.443	11.542	11.447	11.636
2011-12	11.664	11.781	11.687	11.876

Conclusion. The forecasts reproduce the strong annual pattern and the realized 2011 values are broadly tracked by the point forecasts and intervals.

Question 2

Seasonally adjusted WTI crude-oil price

Monthly data in dollars per barrel. The training sample ends in December 2022; 2023M01–2024M10 is held out.

(a) Oil price in levels

Method. Plot the seasonally adjusted price level over the complete sample.

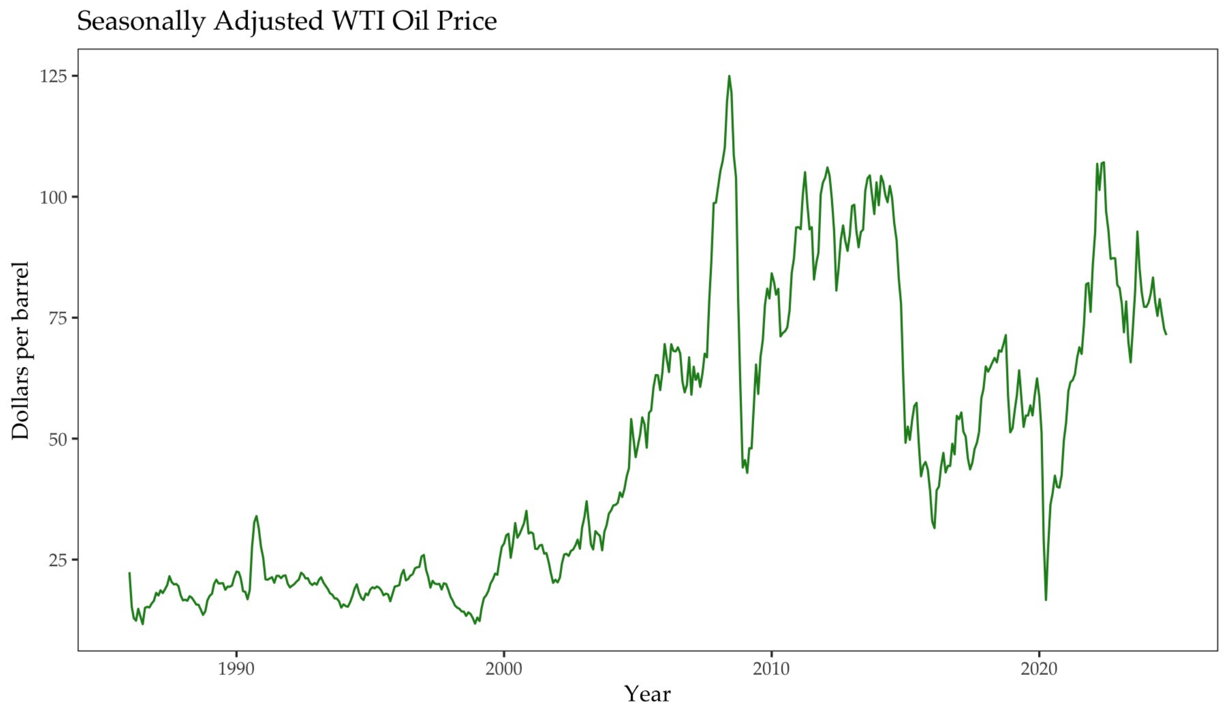

Figure 8. Monthly seasonally adjusted WTI oil price.

Interpretation. The level has prolonged rises and falls, major breaks, and a changing mean. Shocks appear persistent rather than quickly mean-reverting.

Conclusion. The level does not look stationary.

(b) ACF and PACF of the level

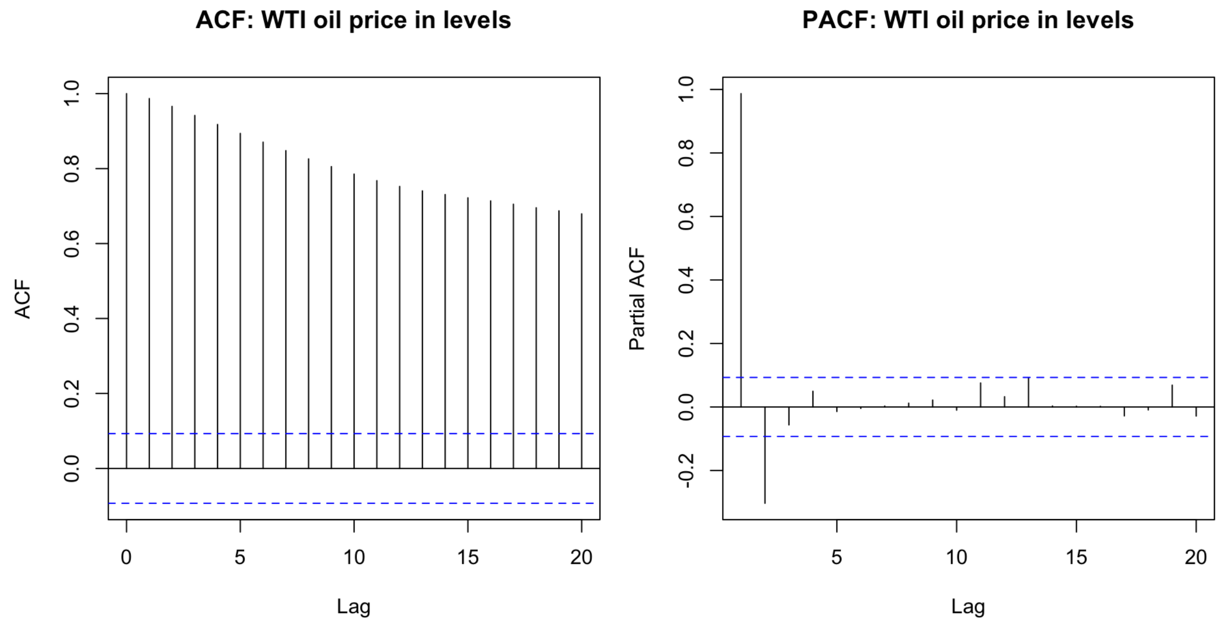

Figure 9. ACF and PACF of the WTI oil-price level.

Interpretation. The ACF begins at 0.987 at lag 1 and decays very slowly, while the PACF is dominated by the first lag. This pattern is consistent with a highly persistent, possibly unit-root process.

Conclusion. Formal Dickey–Fuller tests are required before modeling the level.

(c) Manual Dickey–Fuller regressions

Method. Estimate Δy_t on y_{t-1} under the baseline, drift, and drift-plus-trend specifications. The DF statistic is the t statistic on y_{t-1} and must be compared with nonstandard DF critical values.

Specification	$\hat{\varphi}$	Std. error	DF statistic
Baseline	-7.692e-04	0.003756	-0.205
Drift	-0.009836	0.006976	-1.410
Drift and trend	-0.023463	0.010156	-2.310

Interpretation. The statistics (−0.205, −1.410, and −2.310) are not sufficiently negative to reject a unit root under their respective DF distributions.

Conclusion. The manual DF regressions do not reject a unit root in the oil-price level.

(d) ADF test from aTSA

Specification	Lag	ADF	p-value
No drift / no trend	0	-0.205	0.585
Drift	0	-1.410	0.556
Drift + trend	0	-2.310	0.445

Interpretation. At lag 0, aTSA reproduces the three manual statistics. The p-values are 0.585, 0.556, and 0.445, so the null of a unit root is not rejected. The additional reported lag choices lead to the same substantive conclusion at the 5 percent level.

Conclusion. The package output agrees with the manual DF calculations.

(e) First differences and order of integration

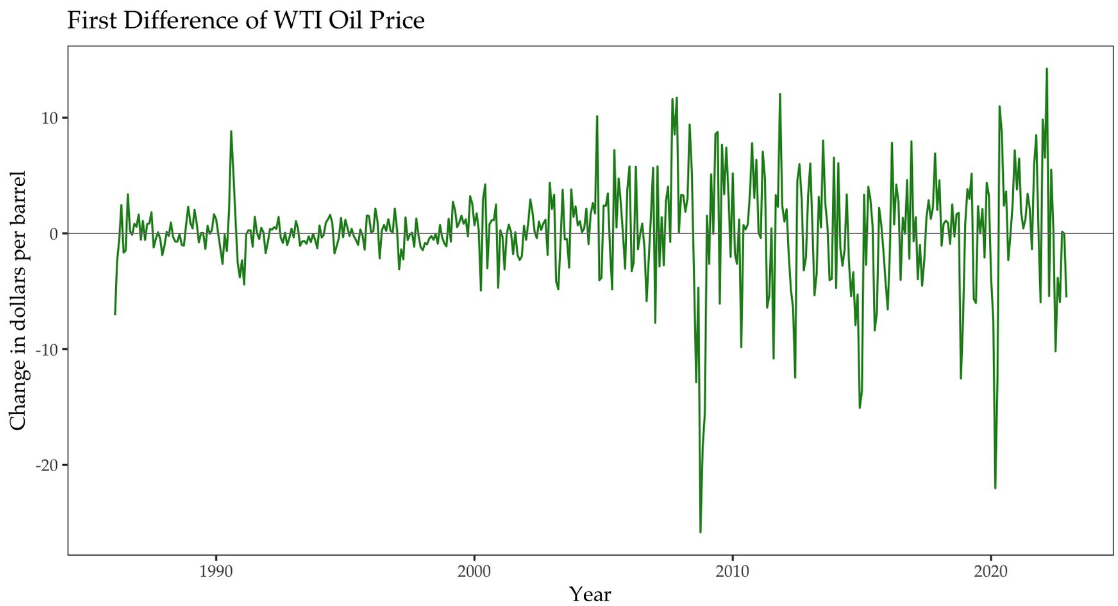

Figure 10. Monthly change in WTI oil price.

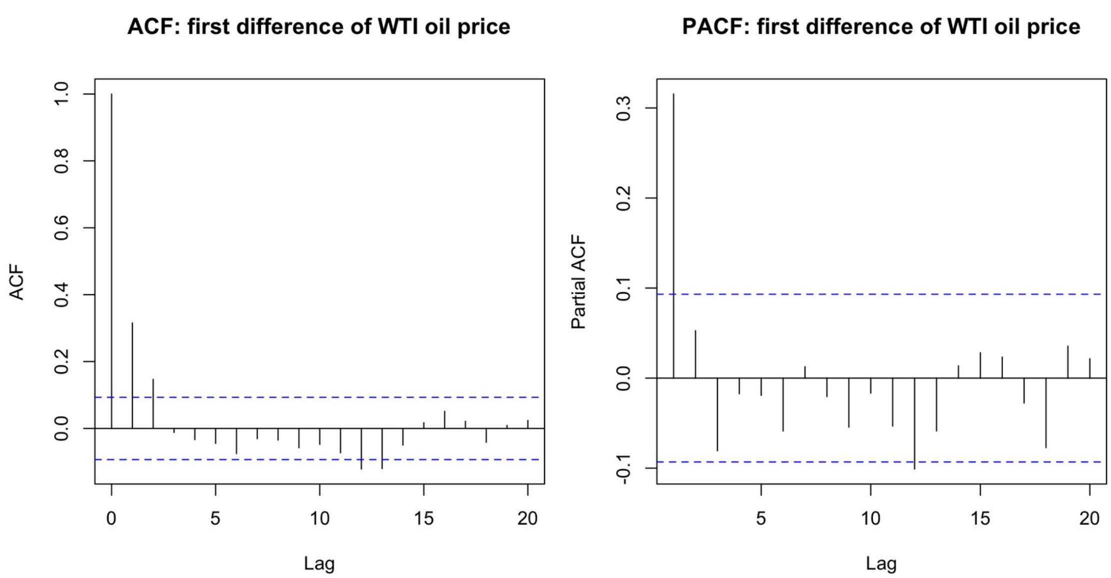

Figure 11. ACF and PACF of the first difference.

Specification	ADF	p-value
No drift / no trend	-15.13	≤ 0.01
Drift	-15.13	≤ 0.01

Specification	ADF	p-value
Drift + trend	-15.11	≤ 0.01

Interpretation. Differencing removes the slow decay. The change series fluctuates around zero; only a few short-lag ACF spikes remain. All three lag-0 ADF statistics are about −15.1 with p-values no larger than 0.01.

Conclusion. Δy_t is stationary and the oil-price level is consistent with an I(1) process.

(f) ARMA model for monthly oil-price changes

AR order	MA(0)	MA(1)	MA(2)	MA(3)
AR(0)	2562.09	2530.05	2523.50	2529.59
AR(1)	2521.32	2526.62	2529.59	2528.37
AR(2)	2526.19	2530.96	2535.68	2541.76
AR(3)	2529.35	2530.18	2535.56	2540.31

Selected model	Parameter	Estimate	Std. error
ARMA(1,0)	ar1	0.3181	0.0452
ARMA(1,0)	intercept	0.1202	0.2840

Interpretation. ARMA(1,0) has the lowest BIC, 2521.32. Its residual ACF has no significant spikes through lag 20, so it adequately captures the short-run dependence visible in Δy_t.

Conclusion. Use ARMA(1,0) with a constant to forecast monthly changes.

(g) Forecasts of monthly changes, 2023–2024

Method. Generate 22 ARMA(1,0) forecasts and 95 percent intervals for January 2023 through October 2024.

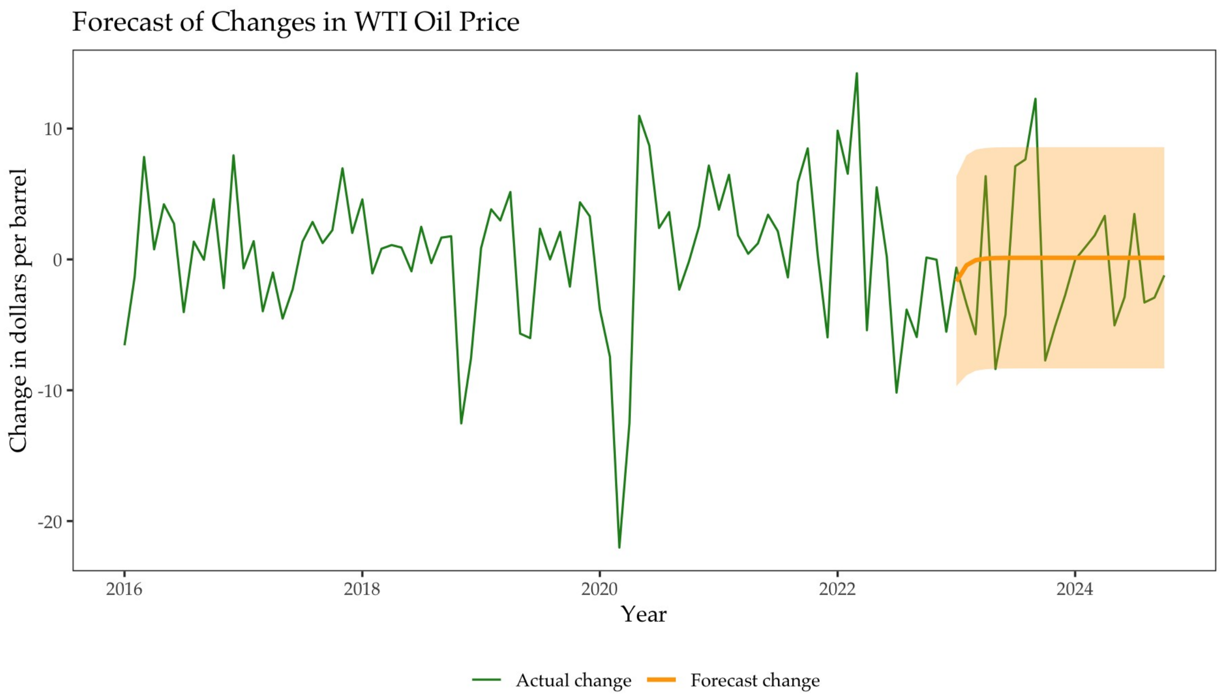

Figure 12. Actual and forecast monthly WTI price changes, 2016–2024.

Month	Actual Δ	Forecast Δ	Lower 95%	Upper 95%
2023-01	-0.63	-1.68	-9.69	6.34
2023-02	-3.41	-0.45	-8.86	7.96
2023-03	-5.72	-0.06	-8.51	8.39
2024-08	-3.29	0.12	-8.33	8.57
2024-09	-2.92	0.12	-8.33	8.57
2024-10	-1.22	0.12	-8.33	8.57

Interpretation. The AR effect decays rapidly, so point forecasts return close to the estimated mean monthly change. The wide intervals appropriately reflect the high month-to-month variability.

(h) Forecasts of the oil-price level, 2023–2024

Method. Use the equivalent ARIMA(1,1,0) representation to obtain level forecasts and valid multi-step intervals. Its point forecasts are cross-checked against the last 2022 price plus cumulative ARMA change forecasts.

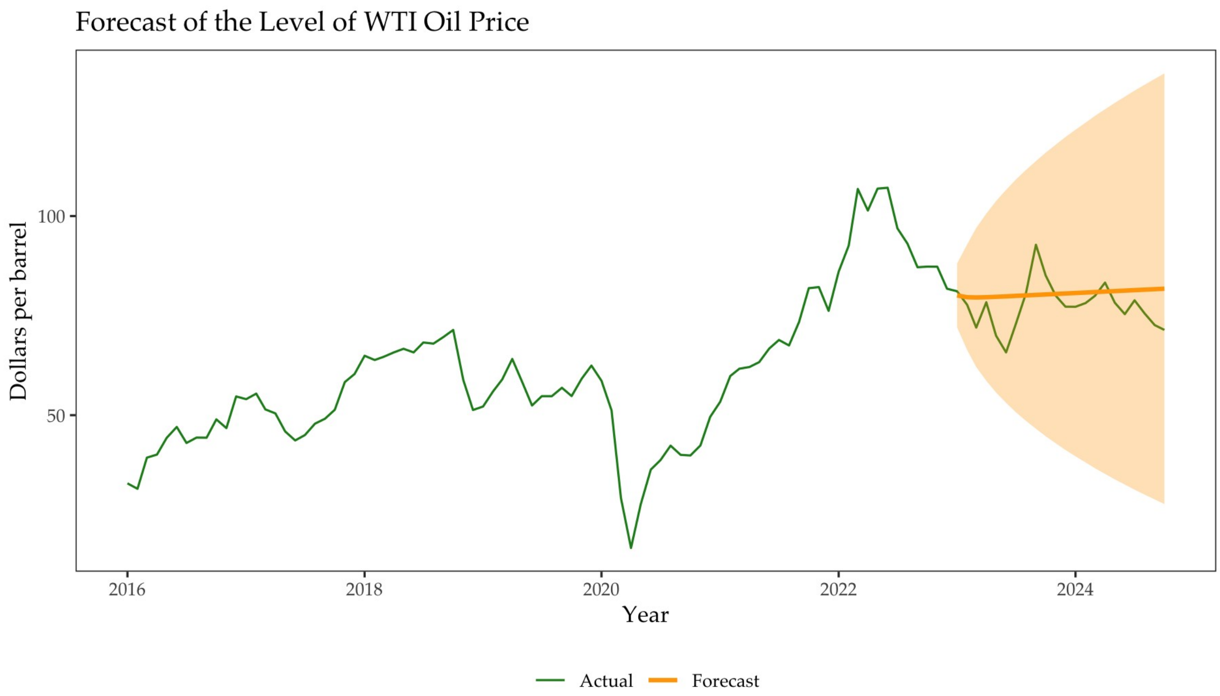

Figure 13. Actual and forecast WTI oil-price level, 2016–2024.

Interpretation. The maximum difference between the two point-forecast constructions is 1.85×10^{-7}, confirming numerical equivalence. The level intervals widen with the horizon because uncertainty accumulates in an I(1) process.

Conclusion. The model produces a nearly flat central level forecast with increasingly uncertain long-horizon outcomes.

Question 3

Disney stock returns and conditional volatility

Daily closing prices. The return is $r_t = \Delta\log(P_t)$; the training sample ends on September 30, 2024.

(a) Log price and stationarity

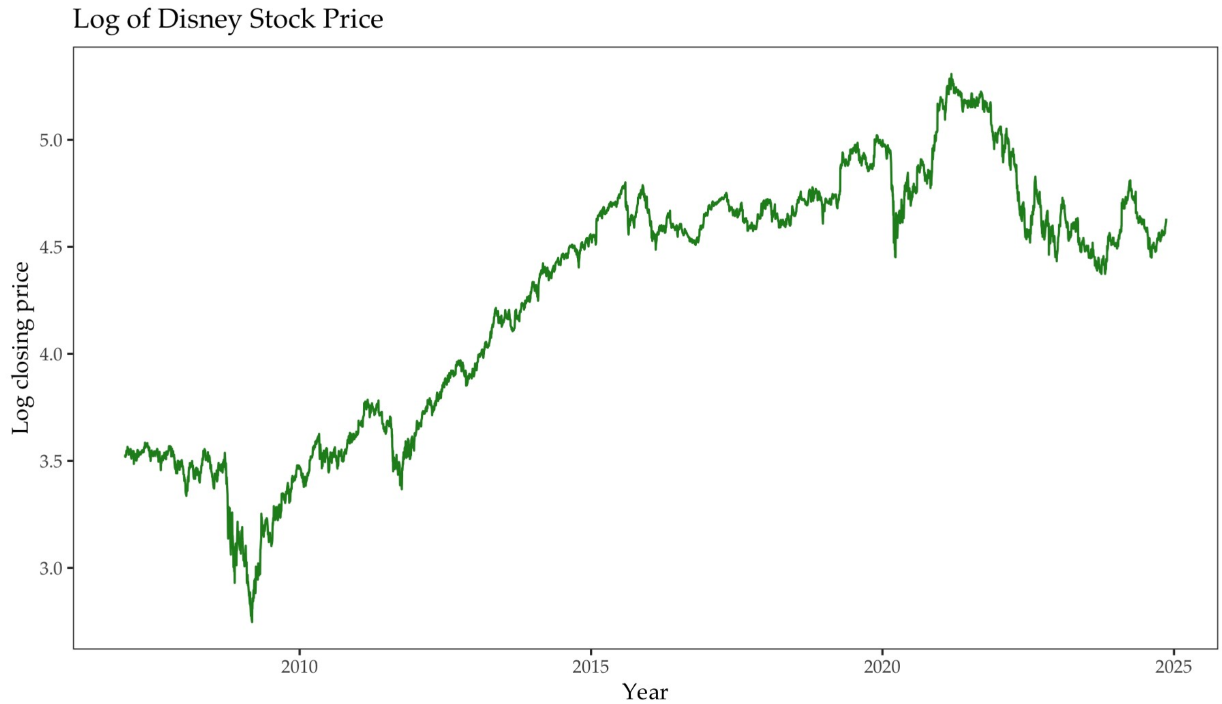

Figure 14. Log Disney closing price.

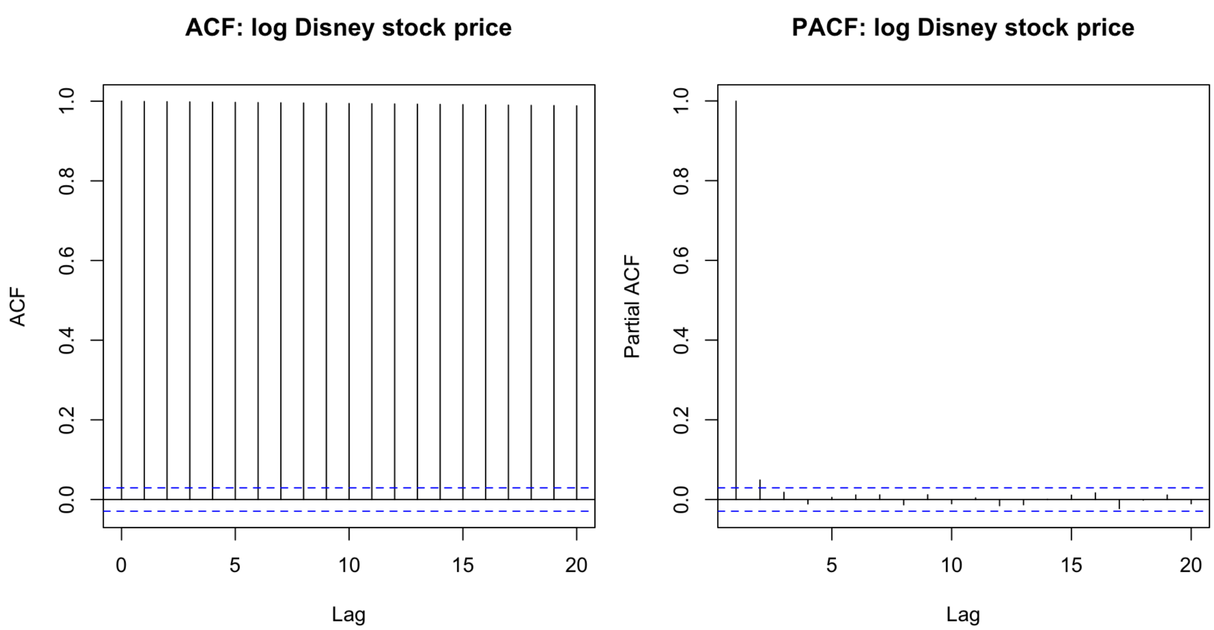

Figure 15. ACF and PACF of the log price.

Interpretation. The log price moves for long periods at different levels and the ACF decays very slowly from values near one. This is the classic visual pattern of a nonstationary price level.

Conclusion. Analyze first differences of the log price rather than the level.

(b) Daily log returns and volatility clustering

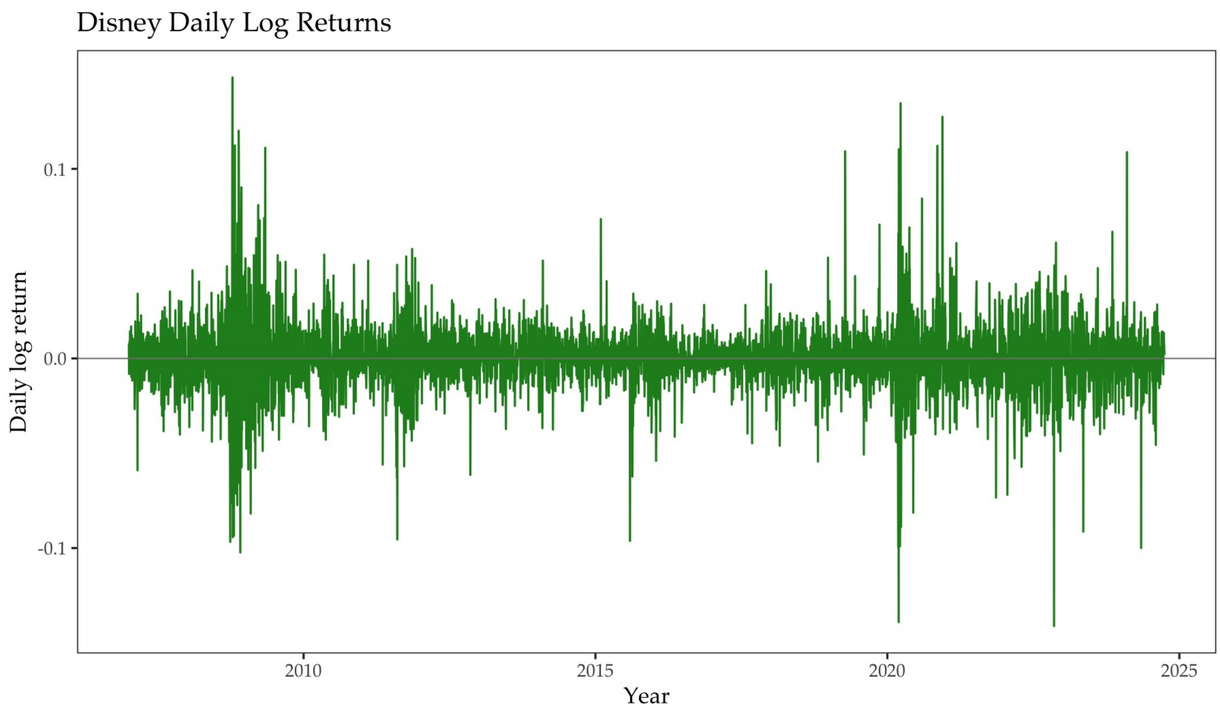

Figure 16. Disney daily log returns.

Interpretation. Large absolute returns occur in clusters rather than independently. Volatility is especially high during the 2008–2009 financial crisis and the 2020 pandemic episode, with additional bursts in 2011, 2015–2016, and 2022–2024.

Conclusion. The return series exhibits volatility clustering and motivates ARCH/GARCH modeling.

(c) Distribution and descriptive statistics

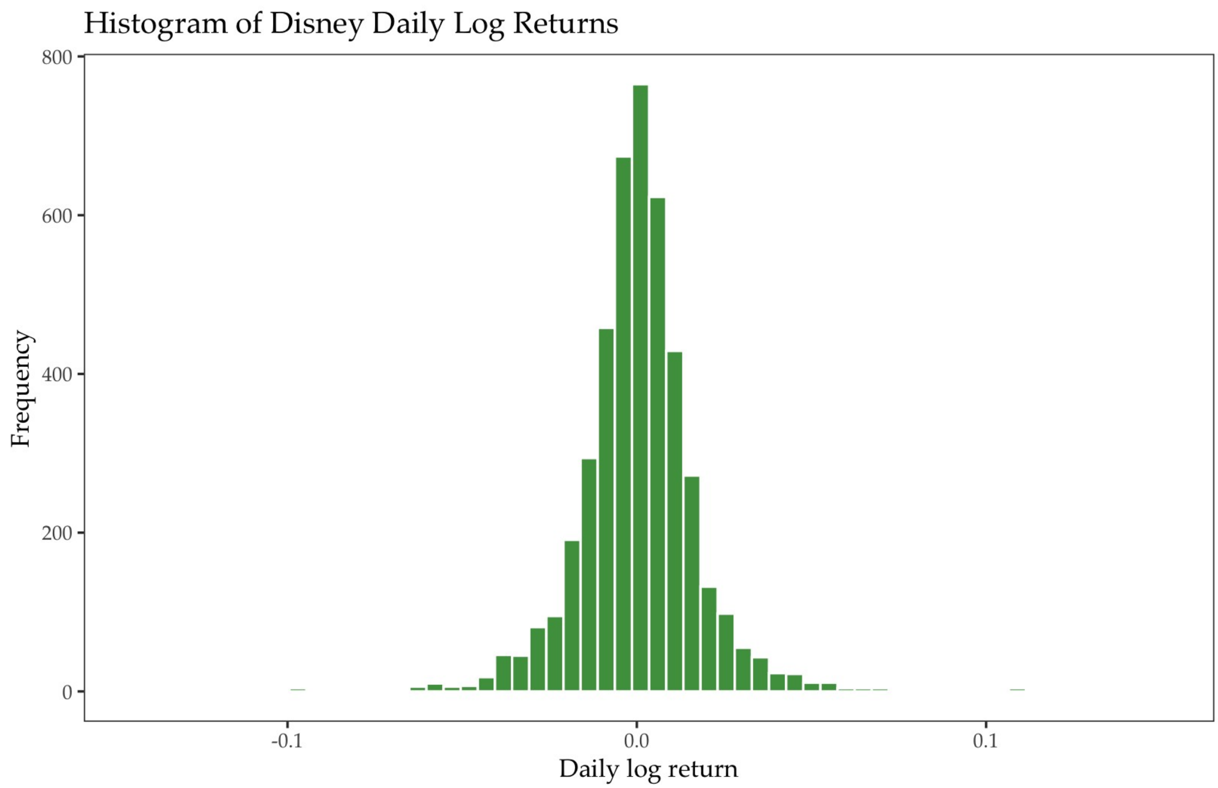

Figure 17. Histogram of Disney daily log returns.

Statistic	Value
Mean	2.346e-04
Median	3.703e-04
Standard deviation	0.017994
Skewness	0.093850
Kurtosis	12.354
Excess kurtosis	9.354

Interpretation. Mean and median are close to zero and skewness is only mildly positive. Ordinary kurtosis is 12.354, far above the normal benchmark of 3; excess kurtosis is 9.354. The histogram is sharply peaked with heavy tails.

Conclusion. Disney returns are strongly leptokurtic.

(d) Serial correlation in squared returns

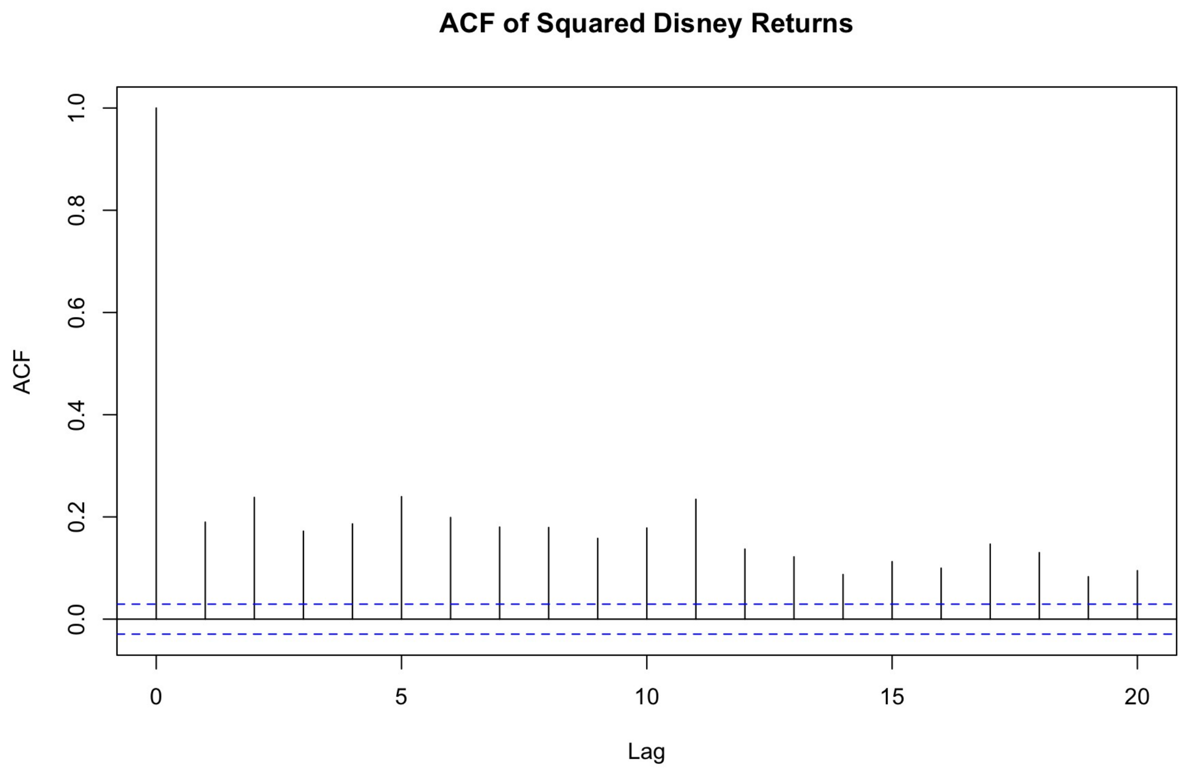

Figure 18. ACF of squared Disney returns through lag 20.

Interpretation. All first 20 autocorrelations are positive and above their approximate 95 percent bounds. The lag-1 ACF is about 0.190, and persistence extends well beyond one day.

Conclusion. Squared returns are serially correlated, providing direct evidence of conditional heteroskedasticity.

(e) AR(1) model for squared returns

Coefficient	Estimate	Std. error	t statistic	p-value
Lagged squared return	0.1898	0.0147	12.91	< 0.001

Interpretation. The AR(1) coefficient is positive and highly significant. A large squared return therefore predicts a larger squared return on the next trading day.

Conclusion. Return variance is persistent and time-varying.

(f)–(g) ARCH/GARCH model comparison

Method. Estimate the four models requested in the project under a conditional normal distribution using the course fGarch specification, then select the lowest BIC.

Model	BIC	Finite estimates	Optimizer diagnostic
ARCH(1)	-5.297	Yes	singular convergence (7)
GARCH(1,1)	-5.488	Yes	singular convergence (7)
AR(1)-ARCH(1)	-5.298	Yes	singular convergence (7)
AR(1)-GARCH(1,1)	-5.487	Yes	singular convergence (7)

Model	Parameter	Estimate	Std. error	p-value
ARCH(1)	mu	4.007e-04	2.403e-04	0.095
ARCH(1)	omega	2.303e-04	6.207e-06	<0.001
ARCH(1)	alpha1	0.309254	0.026535	<0.001
GARCH(1,1)	mu	3.182e-04	2.077e-04	0.125
GARCH(1,1)	omega	6.120e-06	1.024e-06	<0.001
GARCH(1,1)	alpha1	0.077967	0.008979	<0.001
GARCH(1,1)	beta1	0.903994	0.010316	<0.001
AR(1)-ARCH(1)	mu	4.150e-04	2.420e-04	0.086
AR(1)-ARCH(1)	ar1	-0.062030	0.019970	0.002
AR(1)-ARCH(1)	omega	2.269e-04	6.357e-06	<0.001
AR(1)-ARCH(1)	alpha1	0.329051	0.029283	<0.001
AR(1)-GARCH(1,1)	mu	3.295e-04	2.081e-04	0.113
AR(1)-GARCH(1,1)	ar1	-0.036486	0.016608	0.028
AR(1)-GARCH(1,1)	omega	6.060e-06	1.021e-06	<0.001
AR(1)-GARCH(1,1)	alpha1	0.077933	0.009035	<0.001
AR(1)-GARCH(1,1)	beta1	0.904260	0.010363	<0.001

Interpretation. Within part (f), GARCH(1,1) has a much lower BIC than ARCH(1). Within part (g), AR(1)-GARCH(1,1) beats AR(1)-ARCH(1). Across all four, GARCH(1,1) has the lowest BIC (−5.488), narrowly below AR(1)-GARCH(1,1) (−5.487). In the selected model, $\alpha_1 + \beta_1 = 0.982$, indicating highly persistent conditional variance.

Conclusion. Choose GARCH(1,1) as the best-fitting model by BIC.

> ***Reproducibility note. With the installed fGarch version, the default course fits return finite coefficients, Hessian standard errors, BIC values, and conditional volatilities, but the optimizer reports 'singular convergence (7)' for all four models. No alternate solver, distribution, or model was introduced.***

(h) Estimated conditional volatility

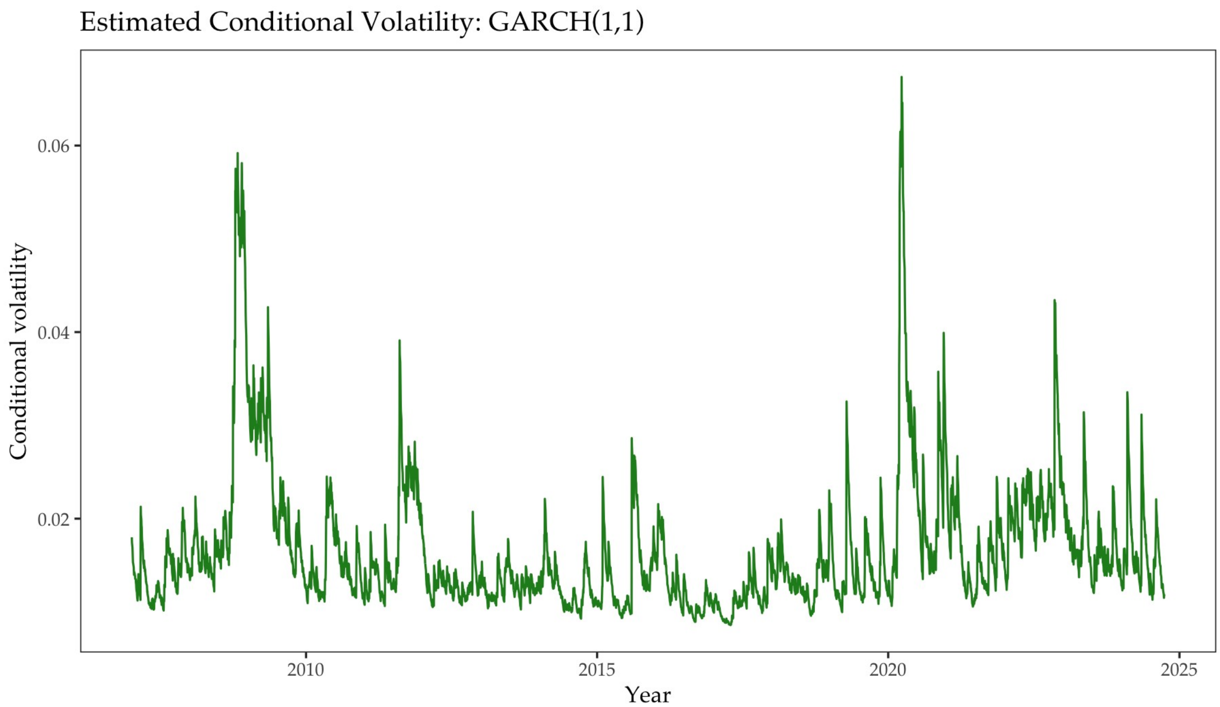

Figure 19. Conditional volatility estimated by the selected GARCH(1,1) model.

Interpretation. Estimated volatility changes substantially over time and decays gradually after large shocks. The largest spikes occur around the 2008–2009 crisis and the 2020 pandemic episode, with the 2020 peak slightly higher, followed by several smaller later episodes.

Conclusion. The GARCH model captures both volatility clustering and persistence.

(i) Conditional-volatility forecasts for October–November 2024

Method. Generate one forecast for each of the 32 observed trading dates from October 1 through November 13, 2024, and align the forecasts with those dates.

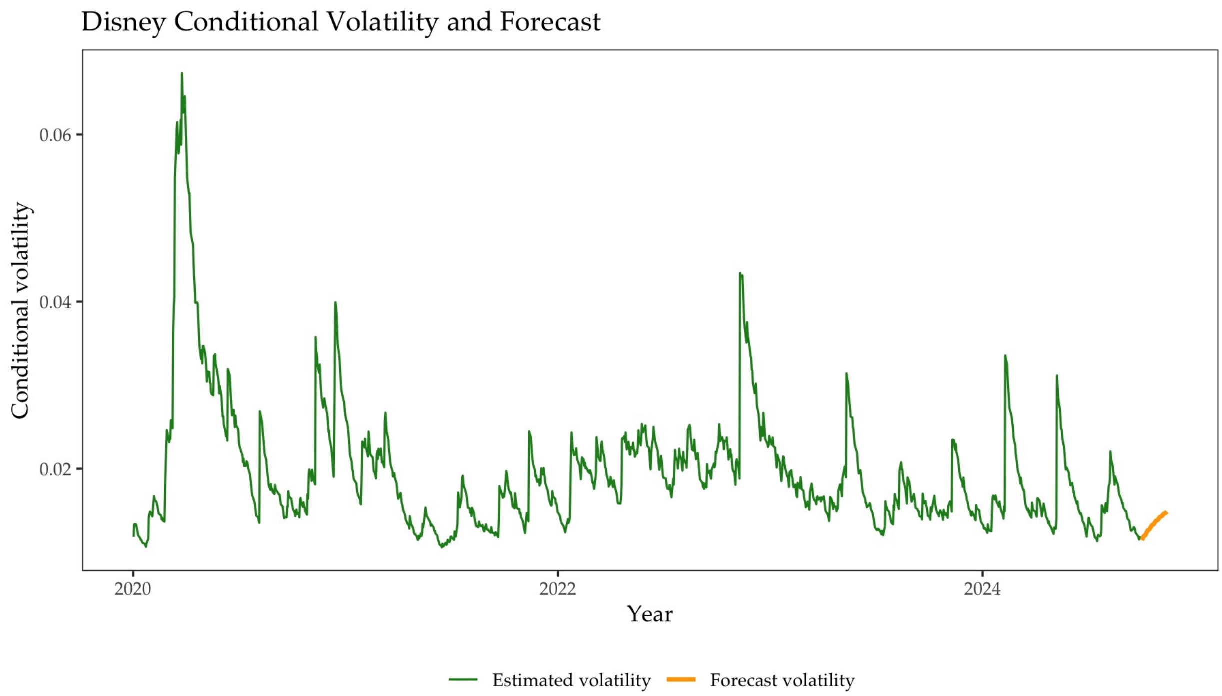

Figure 20. Estimated conditional volatility, 2020–September 2024, and forecasts through November 13, 2024.

Trading date	Volatility forecast
2024-10-01	0.01143
2024-10-02	0.01159
2024-10-03	0.01175
2024-10-04	0.01191
2024-11-08	0.01463
2024-11-11	0.01470
2024-11-12	0.01478
2024-11-13	0.01485

Interpretation. The forecast begins near the late-September volatility level and then gradually moves upward toward the model's long-run conditional volatility as the horizon increases.

Conclusion. The selected GARCH(1,1) model forecasts a smooth, persistent volatility path over the October–November holdout period.

Reproducibility checklist

Automated check	Value
Electricity observations	468
WTI observations	466
Disney observations	4498
Q1 candidate BIC values	15
Q2 candidate BIC values	16
Q1 forecast observations	12
Q2 forecast observations	22
Q3 volatility forecast observations	32
Q2 level-equivalence maximum difference	0.000000184676039793885

Run status. The complete R script runs from a clean session in the Final Project directory without errors and recreates every file used in this report.

Submission pair. Submit this Word document together with ECON475_FinalProject.R. The three source CSV files remain in the project directory for replication.